

Arrhythmia — Clinical Reference Summary

General Recognition and Management Principles

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1. General Classification

Arrhythmias are broadly categorized by origin (supraventricular versus ventricular), rate (bradyarrhythmia versus tachyarrhythmia), and mechanism (automaticity, triggered activity, or re-entry).

Category	Examples
Supraventricular tachyarrhythmias	Atrial fibrillation, atrial flutter, AVNRT, AVRT, focal atrial tachycardia
Ventricular tachyarrhythmias	Monomorphic VT, polymorphic VT, ventricular fibrillation
Bradyarrhythmias	Sinus node dysfunction, AV block (1st, 2nd, 3rd degree)

2. Initial Evaluation

- 12-lead ECG during the arrhythmia when feasible, plus a baseline sinus-rhythm ECG.
- Ambulatory monitoring (Holter, event recorder, or implantable loop recorder) for intermittent symptoms.
- Echocardiography to assess structural heart disease, an important determinant of risk and management.
- Laboratory assessment including electrolytes and thyroid function, given their arrhythmogenic potential.
- Electrophysiologic study in selected patients for mechanism clarification or pre-ablation planning.

3. Supraventricular Tachycardia — General Management

Acute management of regular narrow-complex SVT typically begins with vagal maneuvers followed by intravenous adenosine if maneuvers are unsuccessful, followed by AV nodal blocking agents (beta-blockers, non-dihydropyridine calcium channel blockers) for recurrent or refractory cases. Catheter ablation is a well-established, generally curative option for recurrent AVNRT and accessory-pathway-mediated tachycardia.

4. Ventricular Arrhythmias — General Management

4.1 Sustained Monomorphic VT

Management depends heavily on hemodynamic stability and underlying structural heart disease. Unstable VT requires immediate synchronized cardioversion. Stable, sustained monomorphic VT in the setting of structural

heart disease is often addressed with antiarrhythmic therapy (e.g., amiodarone) and evaluation for ICD implantation and/or catheter ablation.

4.2 Polymorphic VT / Ventricular Fibrillation

These require immediate defibrillation per advanced cardiac life support protocols. Reversible causes (ischemia, electrolyte derangement, QT-prolonging drugs) should be identified and corrected, and ICD implantation is generally considered for survivors without a clearly reversible cause.

5. Bradyarrhythmia — General Management

Symptomatic sinus node dysfunction or high-grade/complete AV block is generally an indication for permanent pacemaker implantation once reversible causes (medication effect, electrolyte disturbance, ischemia) have been excluded or addressed.

6. Risk Stratification for Sudden Cardiac Death

- Left ventricular ejection fraction remains a primary risk marker used in ICD eligibility decisions.
- Specific inherited conditions (e.g., long QT syndrome, hypertrophic cardiomyopathy, Brugada syndrome) carry distinct, condition-specific risk-stratification schemes.
- Family history of sudden cardiac death and syncope of unclear etiology warrant further structured evaluation, including possible genetic counseling and testing.